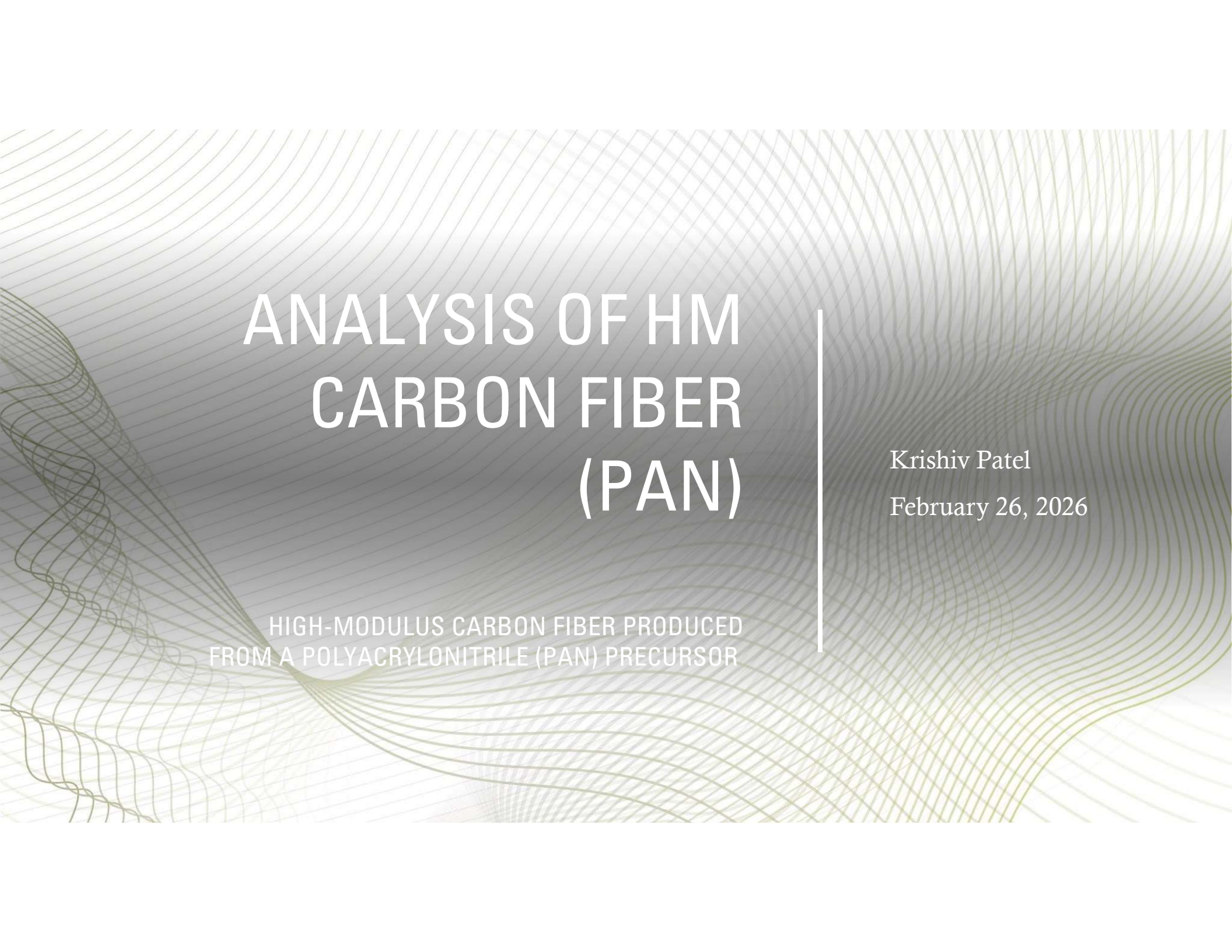




ANALYSIS OF HM CARBON FIBER (PAN)

HIGH-MODULUS CARBON FIBER PRODUCED
FROM A POLYACRYLONITRILE (PAN) PRECURSOR

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OVERVIEW

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INPUT DATA

- Angle..... $\theta = 37.9^\circ$
- Normal Stress (x-plane)..... $\sigma_{xx} = 1,390 \text{ MPa}$
- Normal Stress (y-plane)..... $\sigma_{yy} = 578 \text{ MPa}$
- Shear Stress..... $\tau_{xy} = 496 \text{ MPa}$

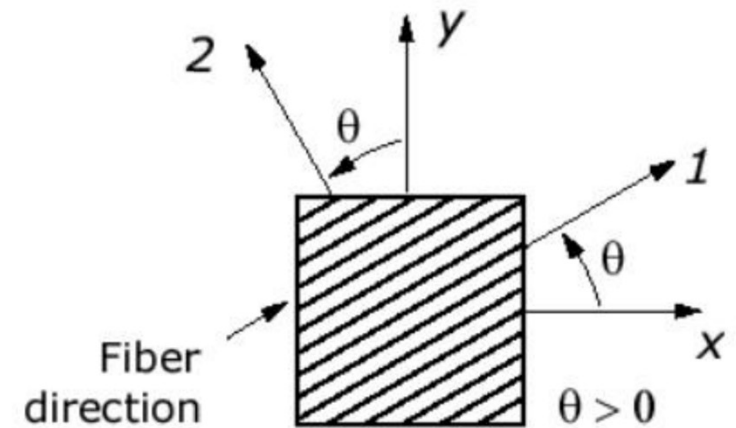
Table 6 Elastic Properties of Unidirectional Polymer Matrix Composites and 7075-T6 Aluminum

Fiber	Axial Modulus GPa	Transverse Modulus GPa	In-Plane Shear Modulus GPa	Major Poisson's Ratio
HM carbon (PAN)	225	7	3.9	.25

```

0 = deg2rad(37.9); %Radians Fiber_angle
o_xx = 1390e6; %Pa Stress_xx
o_yy = 578e6; %Pa Stress_yy
t_xy = 496e6; %Pa Shear

E_11 = 225e9; %Pa. Axial Modulus
E_22 = 7e9; %Pa. Transverse Modulus
G_12 = 3.9e9; %Pa In_Plane Shear Modulus
v_12 = .25; %[no units] Major Poisson Ratio
v_21 = v_12 * E_22 / E_11;
    
```



DELIVERABLES

- Prediction of ϵ_{xx} , ϵ_{yy} , γ_{xy} at given Θ σ_{yy} , τ_{xy}
 - Prediction of ϵ_{11} , ϵ_{22} , γ_{12} at given Θ
 - Prediction of σ_{11} , σ_{22} , τ_{12} at given Θ
 - Charts of ϵ_{11} , ϵ_{22} , γ_{12} and ϵ_{xx} , ϵ_{yy} , γ_{xy} as functions of Θ ($0 < \dots < 90$)
 - Prediction of Min and Max values of ϵ_{11} , ϵ_{22} , γ_{12} and ϵ_{xx} , ϵ_{yy} , γ_{xy}
 - Charts of σ_{11} , σ_{22} , τ_{12} as functions of Θ ($0 < \dots < 90$)
 - Prediction of Min and Max values of σ_{11} , σ_{22} , τ_{12}
-

ALGORITHMS OF ANALYSIS

Variant 1

- Input of $[\sigma_x]$ in non-material orientation
- Calculation of $[\sigma_l]$ in material orientation
- Calculation of $[\epsilon_l]$ in material orientation
- Calculation of $[\epsilon_x]$ in non-material orientation

Variant 2

- Input of $[\sigma_x]$ in non-material orientation
- Calculation of $[S_x]$ in non-material orientation
- Calculation of $[\epsilon]$ in non-material orientation
- Calculation of $[\epsilon_l]$ in material orientation

%Variant 1: Stiffness Transformation

%Inverse of T1

InvT1 = inv(T1);

%We will call [A] the Inverse if T1 * C1 and so on

A = InvT1*C1;

Cx = A*T2;

Sx_1 = inv(Cx);

Ex_1 = Sx_1*O_x

E_xx_1 = Ex_1(1,1);

E_yy_1 = Ex_1(2,1);

Y_xy_1 = Ex_1(3,1);

%Variant 2: Stress Transformation

O_1 = T1*O_x;

O11 = O_1(1,1);

O22 = O_1(2,1);

T12 = O_1(3,1);

E_1 = S1*O_1;

E11 = E_1(1,1);

E22 = E_1(2,1);

Y12 = E_1(3,1);

%The control situation will have Direct Transformation with the Compliance
%Matrix

InvT2 = inv(T2);

B = InvT2*S1;

Sx_2 = B*T1;

Ex_2 = Sx_2*O_x

E_xx_2 = Ex_2(1,1);

E_yy_2 = Ex_2(2,1);

Y_xy_2 = Ex_2(3,1);

$$[T_1] = \begin{bmatrix} m^2 & n^2 & 0 & 0 & 0 & 2mn \\ n^2 & m^2 & 0 & 0 & 0 & -2mn \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & m & -n & 0 \\ 0 & 0 & 0 & n & m & 0 \\ -mn & mn & 0 & 0 & 0 & m^2 - n^2 \end{bmatrix}$$

where $m = \cos \theta$ and $n = \sin \theta$

$$[T_2] = \begin{bmatrix} m^2 & n^2 & 0 & 0 & 0 & mn \\ n^2 & m^2 & 0 & 0 & 0 & -mn \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & m & -n & 0 \\ 0 & 0 & 0 & n & m & 0 \\ -2mn & 2mn & 0 & 0 & 0 & m^2 - n^2 \end{bmatrix}$$

$$\begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & 0 & 0 & 0 \\ S_{12} & S_{22} & S_{23} & 0 & 0 & 0 \\ S_{13} & S_{23} & S_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & S_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & S_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{Bmatrix}$$

$$\left\{ \begin{array}{l} S_{11} = \frac{1}{E_1}, \quad S_{12} = \frac{-\nu_{21}}{E_2}, \quad S_{13} = \frac{-\nu_{31}}{E_3} \\ S_{21} = \frac{-\nu_{12}}{E_1}, \quad S_{22} = \frac{1}{E_2}, \quad S_{23} = \frac{-\nu_{32}}{E_3} \\ S_{31} = \frac{-\nu_{13}}{E_1}, \quad S_{32} = \frac{-\nu_{23}}{E_2}, \quad S_{33} = \frac{1}{E_3} \\ S_{44} = \frac{1}{G_{23}}, \quad S_{55} = \frac{1}{G_{13}}, \quad S_{66} = \frac{1}{G_{12}} \end{array} \right.$$

MATLAB – OUTPUT TERMINAL

```
Ex_1 =
    0.0589
    0.0035
   -0.0650

Ex_2 =
    0.0589
    0.0035
   -0.0650

E_xx_1 =
    0.0589

E_xx_2 =
    0.0589

E_yy_1 =
    0.0035

E_yy_2 =
    0.0035

Y_xy_1 =
   -0.0650

Y_xy_2 =
   -0.0650
```

Both stiffness transformation (Variant 1) and stress transformation (Variant 2) produce identical global strains:

$Ex_1 = Ex_2 = [0.0589; 0.0035; -0.0650]$

This confirms the consistency and correctness of the analysis. Thus the values underneath them can be found using either method

```
Summary_Evaluation_37_9_Degrees =
```

```
3×6 table
```

Global_Strains	Global_Values	Material_Strains	Material_Values	Material_Stresses	Stress_Values
"ε _{xx} "	0.058945	"ε ₁₁ "	0.0065047	"σ ₁₁ "	1.5644e+09
"ε _{yy} "	0.0034724	"ε ₂₂ "	0.055913	"σ ₂₂ "	4.0356e+08
"γ _{xy} "	-0.065003	"γ ₁₂ "	-0.069724	"τ ₁₂ "	-2.7192e+08

MATLAB CODE

The code used to find all values/graphs is as found below

```
clear;
clc;

%Material = IM Carbon (PAN) = High Modulus carbon fiber produced
% from a Polyacrylonitrile (PAN) precursor. It is a
% high-performance material used for extreme rigidity and
% strength-to-weight ratios

% Modulus
E_11 = 225e9; %Pa, Axial Modulus
E_22 = 7e9; %Pa, Transverse Modulus
G_12 = 3e9; %Pa, In-Plane Shear Modulus
% Poisson's Ratio
nu_12 = .25; %no units
% Major Poisson Ratio
nu_21 = nu_12 * E_22 / E_11;

% Compliance Matrix
C1 = inv(S1);

%Stiffness Matrix
S1 = inv(C1);

%Transformation Matrix 1
T1 = [(cos(0))^2*(sin(0))^2*2*sin(0)*cos(0); (sin(0))^2*cos(0))^2*-2*sin(0)*cos(0); -sin(0)*cos(0)*sin(0)*cos(0) (cos(0))^2*(sin(0))^2];

%Transformation Matrix 2
T2 = [(cos(0))^2*(sin(0))^2*2*sin(0)*cos(0); (sin(0))^2*cos(0))^2*-2*sin(0)*cos(0); -2*sin(0)*cos(0)*sin(0)*cos(0) (cos(0))^2*(sin(0))^2];

%Variant 1: Stress Transformation

%Inverse of T1
InvT1 = inv(T1);

%We will call [A] the Inverse if T1 * C1 and so on
A = InvT1*C1;

Cx = A*T2;

Sx_1 = inv(Cx);

Ex_1 = Sx_1*0_x;

%Plot Lines
figure(1); % Open a new figure window
hold on; % Hold the current figure
plot(x, y1, 'r-', 'LineWidth', 2); % First line: red solid
plot(x, y2, 'b-', 'LineWidth', 2); % Second line: blue solid
plot(x, y3, 'y-', 'LineWidth', 2); % Third line: yellow solid
hold off; % Release the hold

%Variant 2: Stress Transformation

%Add labels and legend
xlabel('Angle [Degrees]');
ylabel('Stress [Pa]');
title('Material Stresses on IM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads');
legend('sigma_xx', '\gamma_{xy}', '\gamma_{xy}');
grid on; % Optional: add grid lines

% Global strains (y7, y8, y9)
[max_epsxx, idx_max_epsxx] = max(y7);
[min_epsxx, idx_min_epsxx] = min(y7);
[max_epssy, idx_max_epssy] = max(y8);
[min_epssy, idx_min_epssy] = min(y8);
[max_gammaxy, idx_max_gammaxy] = max(y9);
[min_gammaxy, idx_min_gammaxy] = min(y9);

% Convert indices to angles (x is in degrees)
angle_max_sig11 = x(idx_max_sig11);
angle_min_sig11 = x(idx_min_sig11);
angle_max_sig22 = x(idx_max_sig22);
angle_min_sig22 = x(idx_min_sig22);
angle_max_tau12 = x(idx_max_tau12);
angle_min_tau12 = x(idx_min_tau12);

% Plot Lines
figure(2); % Open a new figure window
hold on; % Hold the current figure
plot(x, y5, 'b-', 'LineWidth', 2); % First line: red solid
plot(x, y6, 'y-', 'LineWidth', 2); % Second line: blue solid
plot(x, y7, 'y-', 'LineWidth', 2); % Third line: yellow solid
hold off; % Release the hold

% Add labels and legend
xlabel('Angle [Degrees]');
ylabel('Strain');
title('Material Strains on IM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads');
legend('epsilon_11', '\epsilon_{22}', '\gamma_{12}');
grid on;

% Global strains (y7, y8, y9)
[max_eps11, idx_max_eps11] = max(y7);
[min_eps11, idx_min_eps11] = min(y7);
[max_eps22, idx_max_eps22] = max(y8);
[min_eps22, idx_min_eps22] = min(y8);
[max_gamma12, idx_max_gamma12] = max(y9);
[min_gamma12, idx_min_gamma12] = min(y9);

% Convert indices to angles (x is in degrees)
angle_max_eps11 = x(idx_max_eps11);
angle_min_eps11 = x(idx_min_eps11);
angle_max_eps22 = x(idx_max_eps22);
angle_min_eps22 = x(idx_min_eps22);
angle_max_gamma12 = x(idx_max_gamma12);
angle_min_gamma12 = x(idx_min_gamma12);

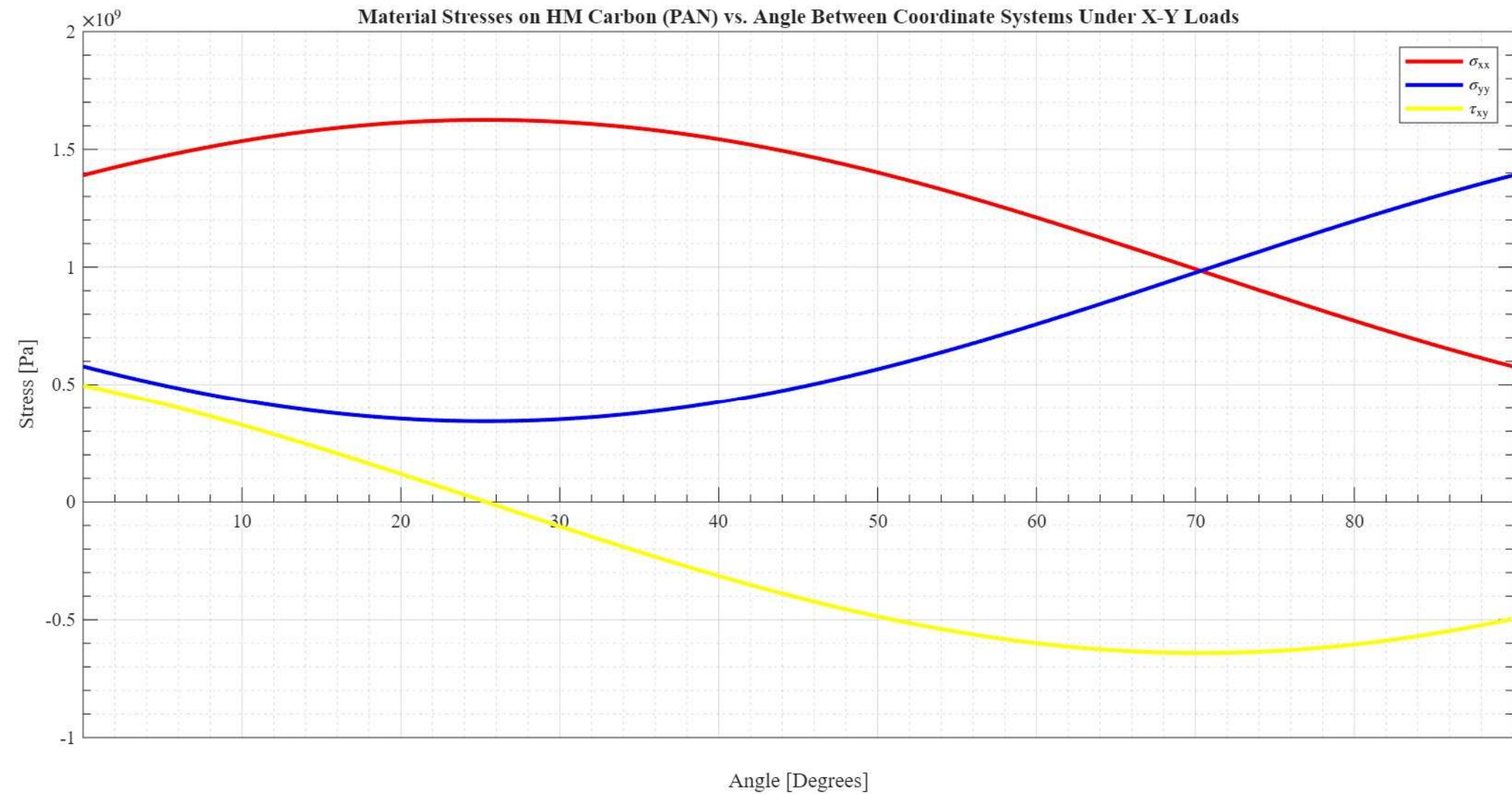
% Plot Lines
figure(3); % Open a new figure window
hold on; % Hold the current figure
plot(x, y7, 'r-', 'LineWidth', 2); % First line: red solid
plot(x, y8, 'b-', 'LineWidth', 2); % Second line: blue solid
plot(x, y9, 'y-', 'LineWidth', 2); % Third line: yellow solid
hold off; % Release the hold

% Add labels and legend
xlabel('Angle [Degrees]');
ylabel('Strain');
title('Global Strain on IM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads');
legend('epsilon_xx', '\epsilon_{yy}', '\gamma_{xy}');
grid on;

% Max and Min Values and Corresponding Angles
% Stresses (y1, y2, y3)
[max_sig11, idx_max_sig11] = max(y1);
[min_sig11, idx_min_sig11] = min(y1);
[max_sig22, idx_max_sig22] = max(y2);
[min_sig22, idx_min_sig22] = min(y2);
[max_tau12, idx_max_tau12] = max(y3);
[min_tau12, idx_min_tau12] = min(y3);

% Material strains (y4, y5, y6)
[max_eps11, idx_max_eps11] = max(y4);
[min_eps11, idx_min_eps11] = min(y4);
[max_eps22, idx_max_eps22] = max(y5);
[min_eps22, idx_min_eps22] = min(y5);
[max_gamma12, idx_max_gamma12] = max(y6);
[min_gamma12, idx_min_gamma12] = min(y6);
```

Material Stresses on HM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads



```

x = 0:0.1:90;

y1 = o_xx * cosd(x).^2 + o_yy * sind(x).^2 + 2 * t_xy * sind(x) .* cosd(x);
y2 = o_xx * sind(x).^2 + o_yy * cosd(x).^2 - 2 * t_xy * sind(x) .* cosd(x);
y3 = -o_xx * sind(x) .* cosd(x) + o_yy * sind(x) .* cosd(x) + t_xy * (cosd(x).^2 - sind(x).^2);

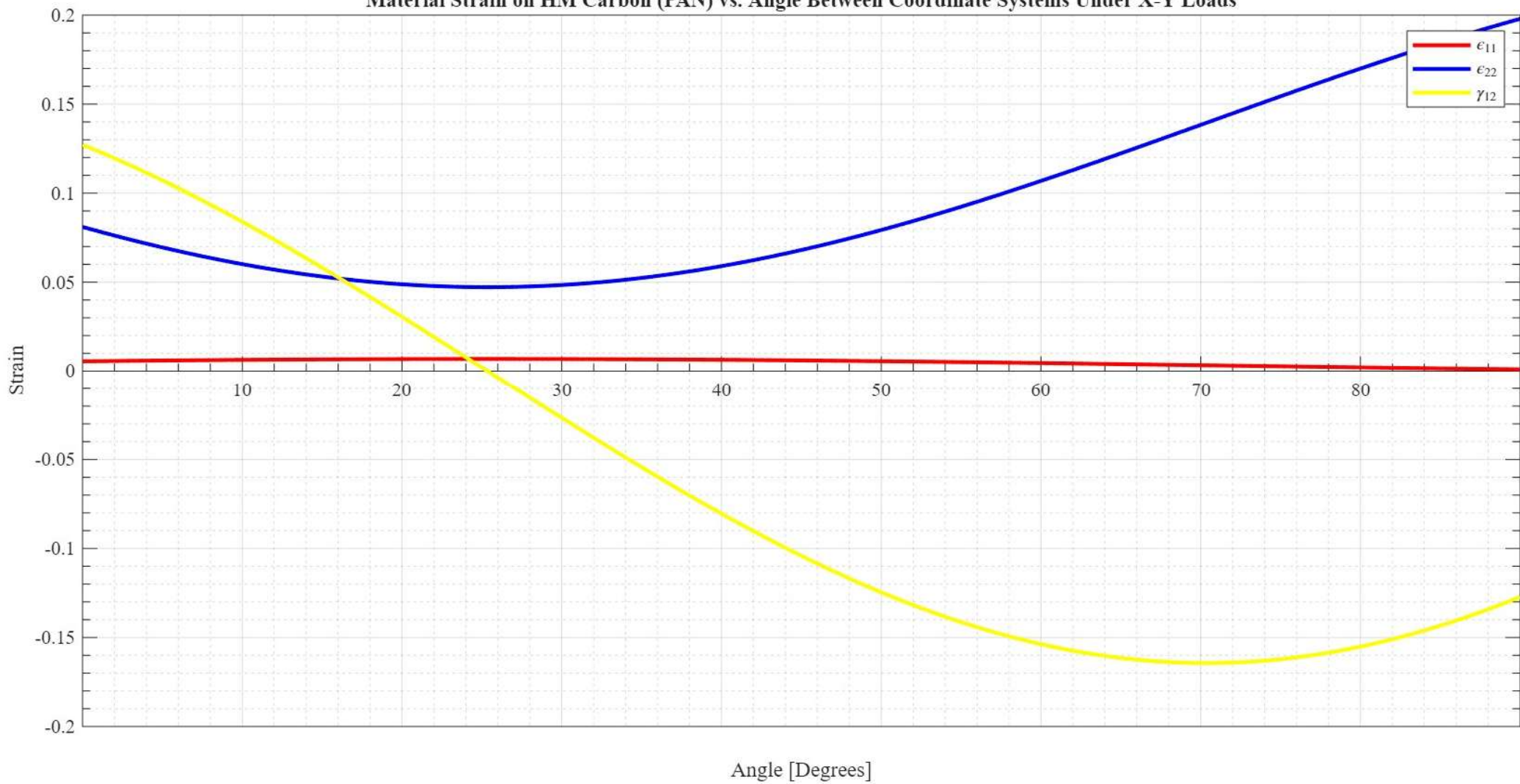
% Plot lines
figure(1); % Open a new figure window
plot(x, y1, 'r-', 'LineWidth', 2); % First line: red solid
hold on; % Hold the current figure
plot(x, y2, 'b-', 'LineWidth', 2); % Second line: blue solid
plot(x, y3, 'y-', 'LineWidth', 2); % Third line: yellow solid
hold off; % Release the hold

% Add labels and legend
xlabel('Angle [Degrees]');
ylabel('Stress [Pa]');
title('Material Stresses on HM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads');
legend('\sigma_{xx}', '\sigma_{yy}', '\tau_{xy}');
grid on; % Optional: add grid lines

```

CODE FOR GRAPH(1)

Material Strain on HM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads



```

y4 = S1(1,1)*y1 + S1(1,2)*y2 + S1(1,3)*y3;
y5 = S1(2,1)*y1 + S1(2,2)*y2 + S1(2,3)*y3;
y6 = S1(3,1)*y1 + S1(3,2)*y2 + S1(3,3)*y3;

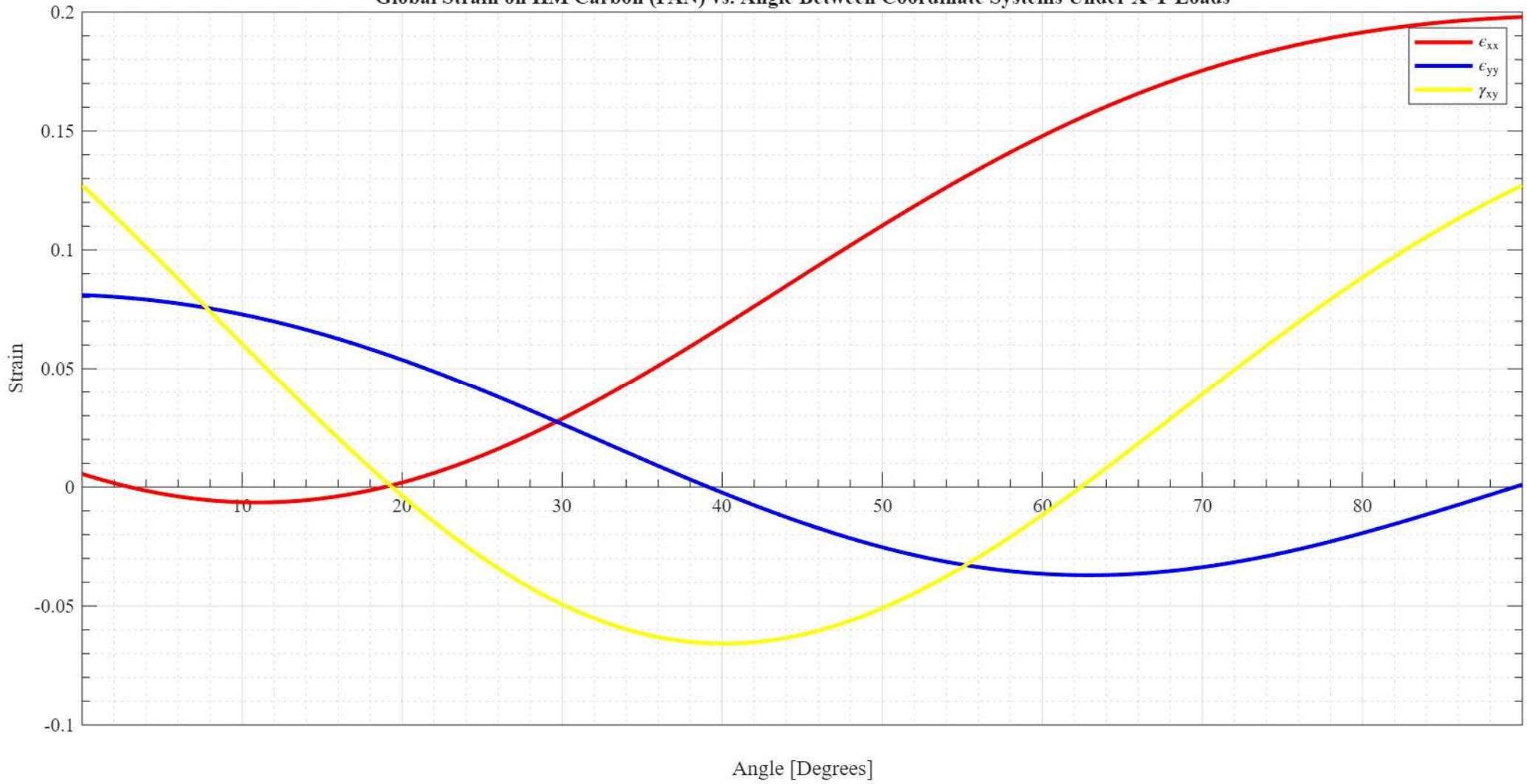
% Plot lines
figure(2); % Open a new figure window
plot(x, y4, 'r-', 'LineWidth', 2); % First line: red solid
hold on; % Hold the current figure
plot(x, y5, 'b-', 'LineWidth', 2); % Second line: blue solid
plot(x, y6, 'y-', 'LineWidth', 2); % Third line: yellow solid
hold off; % Release the hold

xlabel('Angle [Degrees]');
ylabel('Strain');
title('Material Strain on HM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads');
legend('\epsilon_{11}', '\epsilon_{22}', '\gamma_{12}');
grid on;

```

CODE FOR GRAPH(2)

Global Strain on HM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads



```

y7 = y4.*cosd(x).^2 + y5.*sind(x).^2 - y6.*sind(x).*cosd(x);
y8 = y5.*cosd(x).^2 + y4.*sind(x).^2 + y6.*sind(x).*cosd(x);
y9 = 2*sind(x).*cosd(x).*(y4 - y5) + y6.*(cosd(x).^2 - sind(x).^2);
% Plot lines
figure(3); % Open a new figure window
plot(x, y7, 'r-', 'LineWidth', 2); % First line: red solid
hold on; % Hold the current figure
plot(x, y8, 'b-', 'LineWidth', 2); % Second line: blue solid
plot(x, y9, 'y-', 'LineWidth', 2); % Third line: yellow solid
hold off; % Release the hold

xlabel('Angle [Degrees]');
ylabel('Strain');
title('Global Strain on HM Carbon (PAN) vs. Angle Between Coordinate Systems Under X-Y Loads');
legend('\epsilon_{xx}', '\epsilon_{yy}', '\gamma_{xy}');
grid on;

```

CODE FOR GRAPH(3)

MIN/MAX VALUES

===== EXTREME VALUES =====

STRESSES (Pa):

σ_{11}	: max =	1.62e+09	at	25.3°	min =	5.78e+08	at	90.0°
σ_{22}	: max =	1.39e+09	at	90.0°	min =	3.43e+08	at	25.3°
τ_{12}	: max =	4.96e+08	at	0.0°	min =	-6.41e+08	at	70.3°

MATERIAL STRAINS (unitless):

ϵ_{11}	: max =	0.0068	at	25.3°	min =	0.0010	at	90.0°
ϵ_{22}	: max =	0.1979	at	90.0°	min =	0.0472	at	25.3°
γ_{12}	: max =	0.1272	at	0.0°	min =	-0.1644	at	70.3°

GLOBAL STRAINS (unitless):

ϵ_{xx}	: max =	0.1979	at	90.0°	min =	-0.0065	at	11.0°
ϵ_{yy}	: max =	0.0810	at	0.0°	min =	-0.0371	at	62.9°
γ_{xy}	: max =	0.1272	at	0.0°	min =	-0.0658	at	40.1°

CODE FOR MIN/MAX

```
%% Max and Min Values and Corresponding Angles
% Stresses (y1, y2, y3)
[max_sig11, idx_max_sig11] = max(y1);
[min_sig11, idx_min_sig11] = min(y1);
[max_sig22, idx_max_sig22] = max(y2);
[min_sig22, idx_min_sig22] = min(y2);
[max_tau12, idx_max_tau12] = max(y3);
[min_tau12, idx_min_tau12] = min(y3);

% Material strains (y4, y5, y6)
[max_eps11, idx_max_eps11] = max(y4);
[min_eps11, idx_min_eps11] = min(y4);
[max_eps22, idx_max_eps22] = max(y5);
[min_eps22, idx_min_eps22] = min(y5);
[max_gamma12, idx_max_gamma12] = max(y6);
[min_gamma12, idx_min_gamma12] = min(y6);

% Global strains (y7, y8, y9)
[max_epsxx, idx_max_epsxx] = max(y7);
[min_epsxx, idx_min_epsxx] = min(y7);
[max_epsyy, idx_max_epsyy] = max(y8);
[min_epsyy, idx_min_epsyy] = min(y8);
[max_gammaxy, idx_max_gammaxy] = max(y9);
[min_gammaxy, idx_min_gammaxy] = min(y9);

% Convert indices to angles (x is in degrees)
angle_max_sig11 = x(idx_max_sig11);
angle_min_sig11 = x(idx_min_sig11);
angle_max_sig22 = x(idx_max_sig22);
angle_min_sig22 = x(idx_min_sig22);
angle_max_tau12 = x(idx_max_tau12);
angle_min_tau12 = x(idx_min_tau12);

angle_max_eps11 = x(idx_max_eps11);
angle_min_eps11 = x(idx_min_eps11);
angle_max_eps22 = x(idx_max_eps22);
angle_min_eps22 = x(idx_min_eps22);
angle_max_gamma12 = x(idx_max_gamma12);
angle_min_gamma12 = x(idx_min_gamma12);

angle_max_epsxx = x(idx_max_epsxx);
angle_min_epsxx = x(idx_min_epsxx);
angle_max_epsyy = x(idx_max_epsyy);
angle_min_epsyy = x(idx_min_epsyy);
angle_max_gammaxy = x(idx_max_gammaxy);
angle_min_gammaxy = x(idx_min_gammaxy);

%% Display Results
fprintf('\n===== EXTREME VALUES =====\n');
fprintf('STRESSES (Pa):\n');
fprintf('   $\sigma_{11}$  : max = %.2e at %5.1f°    min = %.2e at %5.1f°\n', max_sig11, angle_max_sig11, min_sig11, angle_min_sig11);
fprintf('   $\sigma_{22}$  : max = %.2e at %5.1f°    min = %.2e at %5.1f°\n', max_sig22, angle_max_sig22, min_sig22, angle_min_sig22);
fprintf('   $\tau_{12}$  : max = %.2e at %5.1f°    min = %.2e at %5.1f°\n', max_tau12, angle_max_tau12, min_tau12, angle_min_tau12);

fprintf('\nMATERIAL STRAINS (unitless):\n');
fprintf('   $\epsilon_{11}$  : max = %8.4f at %5.1f°    min = %8.4f at %5.1f°\n', max_eps11, angle_max_eps11, min_eps11, angle_min_eps11);
fprintf('   $\epsilon_{22}$  : max = %8.4f at %5.1f°    min = %8.4f at %5.1f°\n', max_eps22, angle_max_eps22, min_eps22, angle_min_eps22);
fprintf('   $\gamma_{12}$  : max = %8.4f at %5.1f°    min = %8.4f at %5.1f°\n', max_gamma12, angle_max_gamma12, min_gamma12, angle_min_gamma12);

fprintf('\nGLOBAL STRAINS (unitless):\n');
fprintf('   $\epsilon_{xx}$  : max = %8.4f at %5.1f°    min = %8.4f at %5.1f°\n', max_epsxx, angle_max_epsxx, min_epsxx, angle_min_epsxx);
fprintf('   $\epsilon_{yy}$  : max = %8.4f at %5.1f°    min = %8.4f at %5.1f°\n', max_epsyy, angle_max_epsyy, min_epsyy, angle_min_epsyy);
fprintf('   $\gamma_{xy}$  : max = %8.4f at %5.1f°    min = %8.4f at %5.1f°\n', max_gammaxy, angle_max_gammaxy, min_gammaxy, angle_min_gammaxy);
```

CONCLUSIONS - HIGH MODULUS CARBON FIBER (PAN) BEHAVIOR UNDER LOAD

1. **Strong Anisotropy (Stiffness)** Material response depends heavily on fiber orientation under combined loads (σ_{xx} , σ_{yy} , τ_{xy})
2. **Fiber stress (σ_{11})** peaks at **25.3° (1.62 GPa)** – optimal angle to engage fibers for maximum load capacity.
3. **Transverse stress (σ_{22})** is highest at **90° (1.39 GPa)**, indicating matrix-dominated loading off-axis.
4. **Shear stress (τ_{12})** reaches maximum magnitude at **70.3° (−0.64 GPa)**; sign change reflects complex stress state.
5. **Strain behavior:**
 - ϵ_{11} remains low (< 0.7%) due to high axial modulus.
 - ϵ_{22} becomes very large (up to 19.8% at 90°) – risk of matrix failure.
 - γ_{12} is significant (± 0.16), highlighting shear coupling.
6. **Global strains show coupling:** e.g., ϵ_{xx} peaks at **90° (0.198)** while ϵ_{yy} peaks at **0° (0.081)**; γ_{xy} changes sign with orientation.
7. **At 37.9° :**
 - Global: $\epsilon_{xx} \approx 0.059$, $\epsilon_{yy} \approx 0.0035$, $\gamma_{xy} \approx -0.065$
 - Material: $\epsilon_{11} \approx 0.0065$, $\epsilon_{22} \approx 0.056$, $\gamma_{12} \approx -0.070$
 - Stresses: $\sigma_{11} \approx 1.56$ GPa, $\sigma_{22} \approx 0.40$ GPa, $\tau_{12} \approx -0.27$ GPa

Design takeaway: Orient fibers to align with principal stress direction (**~25°**) to maximize load-bearing; avoid **near-90°** to prevent excessive transverse strain.
